

1st Section: general information

1. Innovation Title

(Not more than 5 words)

Bagasse Cushion Wrap

2. New features of Innovation

(Not more than 30 words)

A stretchable cushioning paper made from Thai sugarcane bagasse, using a patterned structure that expands into protective air spaces to absorb impact without plastic bubble wrap or foam and can be stored easily before stretching it.

3. Related to Sustainable Development Goals: SDGs

(You can choose more than 1 answer)

SDG 8 Sustainable Economic Growth

SDG 9 Industry, Innovation and Infrastructure

SDG 12 Responsible Consumption and Production

4. Related to Human Basic Needs

(You can choose more than 1 answer)

Not directly but can be used in transportation of goods, foods, or medical supply.

5. Application of Biomimicry/Local Wisdom)

(Not more than 30 words)

The expandable pattern imitates natural honeycomb structures that provide cushioning with minimal material, while using Thailand's bagasse resource as the primary fiber.

6. Application of Appropriated Technology

(Not more than 30 words)

Uses low-cost fiber processing, paper forming, patterned cutting and simple mechanical testing suitable for community production and gradual industrial scaling.

7. SWOT Analysis/BMC for social enterprise

(Not more than 30 words)

S: Uses bagasse waste, reduces plastic, lightweight, flexible, easy to store and recyclable.

W: May have lower strength and moisture resistance.

O: Demand for sustainable packaging and community-based production.

T: Competition from cheap plastic and foam packaging.

2nd Section: research project information

1. Executive summary or abstract

(Do not more than 100 words)

This project develops a stretchable protective packaging material from sugarcane bagasse, an agricultural by-product widely available in Thailand. Unlike plastic bubble wrap or foam, the material is formed as a patterned paper sheet that expands when stretched, creating a lightweight three-dimensional honeycomb like structure. The sheet can wrap products, reducing impact and product movement during transportation. The research will optimize bagasse fiber composition, sheet thickness, pattern geometry and processing conditions, then evaluate cushioning, compression, tensile strength, biodegradability and cost. The project aims to create a locally produced, scalable and circular packaging solution for small businesses and communities.

2. Rationale

(Show the context of the area and identify the problem and the needs of the area as well as explain the need and importance of research project to address the critical problem or develop the critical capacity and indicate the research questions)

Context

Thailand has a large sugarcane industry and local sugar production that generates significant amounts of sugarcane bagasse. Although bagasse can be used for fuel, animal food, compost and other purposes, there is an opportunity to develop this locally available waste into a higher value material.

Problem

Protective packaging commonly uses plastic bubble wrap, foam. Although these materials provide good protection against impact, they can create large amounts of waste after use. Some are also difficult to recycle or manage.

Needs

The proposed product needs to be lightweight, flexible, strong enough to protect products, and capable of absorbing impact. The material should be designed to reduce waste and support more sustainable and circular production.

Importance

This research is important because it could reduce the amount of waste by providing one valuable way to use bagasse. The development of bagasse cushioning paper could also reduce dependence on plastic and foam packaging and provide a more sustainable alternative of packaging options.

Research Question

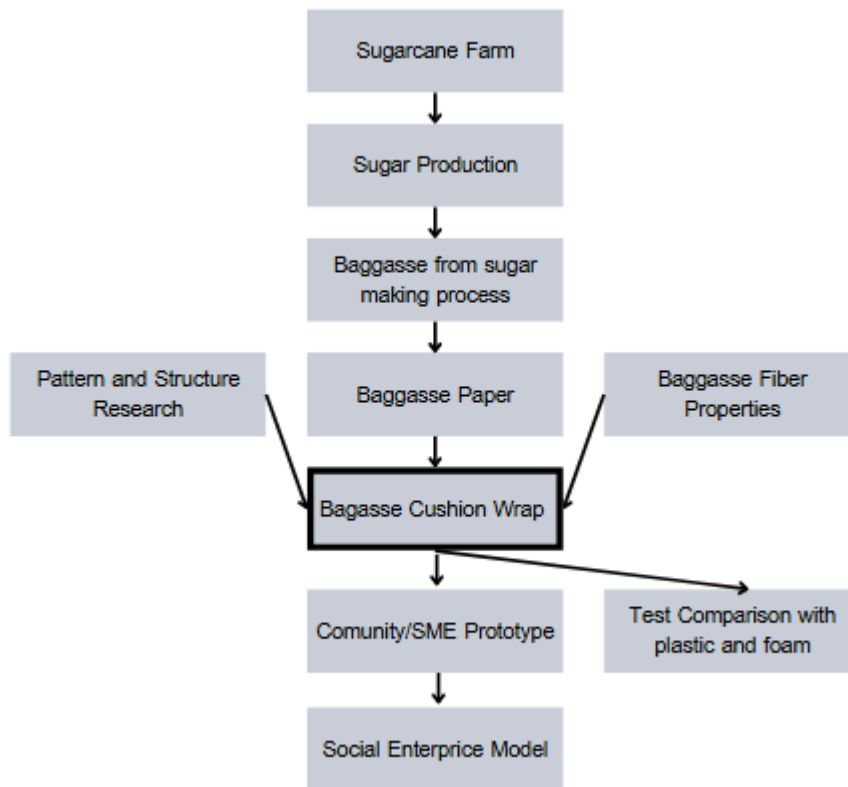
1. What are all the materials and the ration of each for producing the paper?
2. What is the effectiveness of the product compared to plastic and foam?
3. How can this product be implemented in the community scale, small business and or industry scale sugar production.

3. Objectives

(Specify as list)

1. To find and optimize patterned structure for protective packaging.
2. To develop and evaluate strength, resistance and cushioning performance the packaging.
3. To develop a prototype suitable for community and SME scale.
4. To make use of leftovers over bagasse.

4. Research framework and project goals (Please demonstrate your research framework as a schematic diagram to show goals and output/outcome/impact of the project. In addition, the research framework must show collaboration and synergy between both partners)



5. Concepts, theories and research hypotheses

- Circular Economy: system where materials are reused, recycled, or turned into new products instead of thrown away.
- Biomimicry: designing products or solutions by copying nature.
- Sustainable Materials: materials that reduce harm to the environment.

Research Hypotheses

1. Optimized bagasse composition, thickness, and pattern will provide effective product protection.
2. Bagasse cushioning can reduce the need for plastic and foam packaging.
3. Producing bagasse cushioning is possible and opportunities for communities in sugar producing areas.

6. Methodology

1. Material Preparation

Collect, clean, dry, and pulp Thai sugarcane bagasse. Then refine fibers and prepare different paper formulations.

2. Paper Development

Develop sheets with different fiber ratios, lengths, thicknesses, binders, pressing, and drying conditions.

3. Pattern Development

Test patterns such as honeycomb, diamond, hexagonal, slit, and accordion structures.

4. Testing

Evaluate strength, stretchability, compression, impact absorption, vibration, drop performance, and biodegradability.

5. Packaging Prototype

Wrap selected products with the expanded bagasse sheet and place them in cardboard boxes.

6 Comparison

Compare the bagasse wrap with bubble wrap and foam based on protection, weight, cost, usability, and environmental impact.

7 Field Testing

Have community members, SMEs, and packaging users test the prototype and provide feedback on protection, usability, appearance, cost, and acceptance.

8 Social Enterprise Model

Develop a community production, pricing, distribution, and commercialization model.

7. Research sites

(Please specify your research sites - include details such as community/village's name)

Domestic/International	Name of country/province or state	Research sites	Place name
Domestic	Sukhothai	https://eng.kku.ac.th/19056	Tha Chai Village, have done a project on the burnt agricultural waste problem, specifically sugarcane leaves.
Domestic	Khon Kaen	https://www.ksigroup.com/images/download/publications/ar2017-en.pdf	Khon Kaen Sugar Industry, on page 15 have provided usage of sugarcane production waste.
Domestic	Nakhon Sawan	https://www.ktisgroup.com/about/business/product	KTIS Group has used sugar waste to make paper food containers.

8. Benefit sites

Domestic/International	Name of country/province or state	Place name
Domestic	Nakhon Sawan	Pure Plus Farm Community Enterprise
Domestic	-	KEX Express Thailand

9. Specify connection between your project and all your stakeholders

9.1 Connections with other experts within and outside Thailand and planning to build relationships with experts, as well as build future research teams (if any)

- Agricultural waste utilization
- Paper and pulp technology
- Biomaterials
- Packaging engineering
- Mechanical testing

- Product design
- Circular economy
- Social enterprise
- Business commercialization

9.2 Collaborations with stakeholder and user engagement by specifying the names of government, private sector, civil society and community agencies

(Describes the process of working together and the links that drive research results into usefulness. It also describes the ongoing work process of the users taking advantage of the research when the research project is complete.)

Government

- Ministry of Agriculture and Cooperatives
- Ministry of Higher Education, Science, Research and Innovation
- Local government organizations
- Community development agencies

Community development, research support, technology transfer and commercialization.

Private Sector

- Sugar mills
- Packaging manufacturers
- Product manufacturers
- Sustainable-product companies

Civil Society

Environmental organizations and community organizations can support awareness, consumer education and sustainable-packaging adoption.

Community

- Raw-material preparation.
- Prototype production.
- Product testing.
- User feedback.
- Future community enterprise development.

10. Co-funding organization

No.	year	Organization name	Co-operation approach	In-cash (baht)	In-kind	Total
1						
2						

11. Estimated Budget

Duration of project Months

Total budget Baht

Show project budget details divided into different categories as follows

Budget categories	Details	Budget (baht)
Wage		
Material costs		
Expenses		
Equipment		
Total		

12. Expected impacts

Expected impacts	Value propositions deliver	Stakeholder	Use area
Economy			
Social			
Environment			

Appendix 1:

Introductions of innovation

- Why do you feel wow?

The innovation transforms an agricultural residue into a completely different type of packaging material. The user can simply stretch a patterned bagasse-paper sheet and wrap it around the product.

- What will make it more WoW?

- Develop a recognizable Thai-inspired pattern
- Use different natural colors to make the wrap more distinct.
- Make the sheet rollable and easy to store and pull out in a flash.
- Add simple visual indicators showing where to stretch and wrap.

- How can it serve the social enterprise/ sustainable development goals/ biomimicry?

The innovation connects agricultural waste, material innovation and local enterprise. It supports SDG 8 through potential employment, SDG 9 through innovation, and SDG 12 through sustainable resource use and waste reduction.

Biomimicry is inspired by the structure of a beehive, where interconnected hexagonal cells create a lightweight but strong framework. When expanded, these cells increase the material's volume and air pocket, providing cushioning while using relatively little material.

Problem		Pain/Gain Point		Interventions/Innovations/technology	
Plastic and foam waste		Environmental pollution and difficult waste management		Replace with expandable bagasse paper	
Sugarcane bagasse		Underutilized agricultural residue		Convert bagasse into value-added paper	
Product damage		Impact during transport		Expandable structure absorbs force when impact	
Issues	Specific/General	Degree of severity	Impact	Solutions	
Plastic and Foam packaging waste	General	High	Environmental	Nature Base Material	
Agricultural residue utilization	Specific	Medium	Low value resource	Add Value through process	
Product damage during transport	Specific	Medium	Economic	Cushioning Material	
Pain/Gain Points		Appropriated Technology		Expected Outcomes/Changes	
Environmental pollution and difficult waste management cause from the		An Environmental friendly cushion wrap that can be produced from agricultural by-product		Reduced plastic and foam cushioning	

Area Based Canvas

Before Sugarcane bagasse is available as an agricultural by-product that is under utilized Plastic/foam is used for cushioning	Intervention Convert bagasse into paper fiber that can absorb force and can be a more nature friendly alternative	Knowledge/Skills/Ideas proposed Pulping, fiber preparation and paper formulation. Wrap structural design
		Local wisdoms/ Skills/Resources Availability: Sugarcane bagasses
Expected Outcomes/Changes Bagasse resource -> local processing -> expandable cushioning paper ->sustainable packaging ->community/SME product ->circular social enterprise.	After Able to utilize bagasses to become a replaceable option to plastic and foam to use in goods/product transportation as a cushioning in the local sugar communities and in other businesses.	

(CEO)

Appendix 2:

BMC and SWOT analysis
(CEO with CFO/CMO)

Appendix 3:

The Pitch Canvas and The poster/brochure
(CEO with CCO/CIO)

Appendix 4:

Example: The commercialisation strategies
- The social enterprise business strategies: Profits and People (CEO with CFO/CMO)

- The social enterprise business strategies: Planet and People (CEO with CCO/CIO)